

ROSHAN RAJ

Software Systems Engineer & ML Architect

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EDUCATION

Manipal Institute of Technology

Bachelor of Technology — Computer Science Engineering

Bengaluru, India

08/2023 – 08/2027

- Managed research at the Manipal Innovation Cell, incubating early-stage startups.
- Led the attendance-system upgrade for the entire Manipal Institute of Technology Bengaluru campus pilot.

SKILLS

Languages	Python, Swift, TypeScript / JavaScript, SQL, Bash, C/C++
ML / AI	PyTorch, Apple MLX, Hugging Face Transformers, LLM fine-tuning (LoRA, 4-bit quantization), CatBoost, XGBoost, LightGBM, SHAP, scikit-learn, Optuna; statistical evaluation (bootstrap CIs, significance testing)
Quant / Finance	Walk-forward backtesting, feature engineering, weighted ensembles, signal generation, portfolio & margin simulation, pandas / NumPy
Backend & Infra	PostgreSQL, Supabase, Vercel, Node.js, REST APIs, Kubernetes, Docker, serverless, on-premise / air-gapped deployment
Frontend / macOS	Swift, SwiftUI, Metal-accelerated rendering, React, Next.js
Tools	Git, GitHub Actions, Linux, Xcode, Jupyter, Postman

EXPERIENCE

Bastian Build

Co-Founder & Lead Engineer

Bengaluru, India

01/2026 – 04/2026

Infrastructure software company building on-premise (air-gapped), ML-based reliability tooling and authenticated access systems for enterprise.

- Designed an acoustic-handshake authentication protocol resistant to GPS/Bluetooth spoofing; scaled B2B deployment toward 4 university campuses.
- Executed a zero-downtime migration of 31 relational Postgres tables to a serverless Supabase/Vercel stack at 99.9% uptime under institutional traffic.

Lenn Chartered Pvt. Ltd.

Software Engineering Intern

Mumbai, India

12/2025 – 02/2026

AI-driven FinTech education platform deploying domain-specific LLMs and synthetic market simulations.

- Fine-tuned a domain-specific FinTech LLM and built an AI financial advisor on it, reducing client query-resolution time by 90% while maintaining regulatory accuracy.
- Engineered a low-latency synthetic paper-trading engine simulating order execution, routing, and margin for 5,000+ concurrent users.

PROJECTS

LLM Evaluation Framework — Open Source

06/2026

A statistical evaluation and regression-testing framework for LLM applications, applying quantitative-finance evaluation discipline to prompt and model changes.

- Reports every metric with bootstrap (BCa) confidence intervals and paired significance tests, replacing single-number scoring that hides noise.
- Detects evaluation-set leakage and overfitting, and gates CI so changes that statistically regress quality fail the build.
- Supports 4 model backends (OpenAI, Anthropic, Ollama, Apple MLX) with fully-offline air-gapped operation; ships a pytest plugin and a GitHub Action, with async evaluation across 1,000+ cases per run.

Finance OS — Financial Command Center

05/2026

A native macOS application for predictive capital allocation and real-time liquidity tracking, GPU-accelerated to a locked 120fps.

- Built the interface in Swift with Metal-backed rendering locked at 120fps across 5+ liquidity-tracking modules.
- Implemented a predictive cash-flow engine projecting liquidity drawdowns up to 90 days ahead with real-time state updates.

Acoustic AMS — Institutional Scale

05/2026

A high-availability attendance system with strict device authentication, using an acoustic-handshake protocol to prevent proximity spoofing.

- Deployed campus-wide at Manipal Institute of Technology Bengaluru at 99.9% uptime and sub-50ms verification latency, scaling toward 4–5 campuses.
- Designed the acoustic-handshake protocol to resist GPS and Bluetooth relay/spoofing for proximity-based check-in.

On-Premise Site Reliability Engine

04/2026

An air-gapped reliability platform running a 14B-parameter LLM locally to diagnose and auto-remediate enterprise server failures with no outbound traffic.

- Engineered an on-device inference pipeline running a 14B-parameter, 4-bit-quantized LLM on Apple MLX with 0 bytes of outbound network traffic.
- Built a digital-twin sandbox that detects microservice anomalies and auto-generates Kubernetes/Bash remediation scripts in under 800ms; tracks real-time outage burn rate (e.g., \$10,000+/min) on a 120fps GPU-accelerated frontend.

Quantitative Signal Engine — ML Trading Research

03/2026

A machine-learning quantitative trading pipeline using weighted gradient-boosting ensembles for low-latency signal generation.

- Built a multi-model pipeline trained on 4,450,146 historical market rows across a 160-day walk-forward validation window.
- Deployed a Nelder-Mead-optimized ensemble (CatBoost / LightGBM / XGBoost) reaching a 54.7% peak win rate at a 0.75 confidence threshold; pruned 36 → 30 features via SHAP across 175 Optuna trials.

PATENTS

Named inventor on 7 patent applications (submitted for filing), spanning privacy, distributed systems, ML infrastructure, security, and embedded hardware:

- Privacy-Preserving Crash Reproduction** — reproduces production crashes from path-equivalent synthetic inputs, with zero bytes of user data transmitted.
- Semantic Clipboard** — paste-time program synthesis transforming copied data into the destination's schema, with live, refreshable provenance links.
- Negative-Latency Collaborative Editing** — speculatively applies predicted remote edits under a formally bounded-divergence rollback guarantee.
- On-Device Pre-Update Regression Certification** — privately replays a user's real workflows to certify an update before install, no data leaving the device.
- Unified Fidelity-Budget Scheduler** — allocates one end-to-end error budget across precision, compression, and sampling knobs with a certified accuracy bound.
- Acoustic Proximity Authentication** — acoustic-handshake protocol resistant to GPS/Bluetooth spoofing for proximity-based identity verification.
- Dynamically Morphing Ergonomic Keyboard** — motorized input device that adapts its geometry in real time from on-device biometric (thermal/force) telemetry.

AWARDS

Flinders AI Hackathon — Top 5 Finalist

Ranked top 5 of 1,000+ engineers in Bengaluru; architected, deployed, and pitched a production-ready AI prototype to global stakeholders under time constraints.

Hackbricks by Databricks — 2nd Runner-Up

Podium finish in a Databricks-sponsored enterprise event; engineered a scalable data solution aligned to live corporate problem statements.